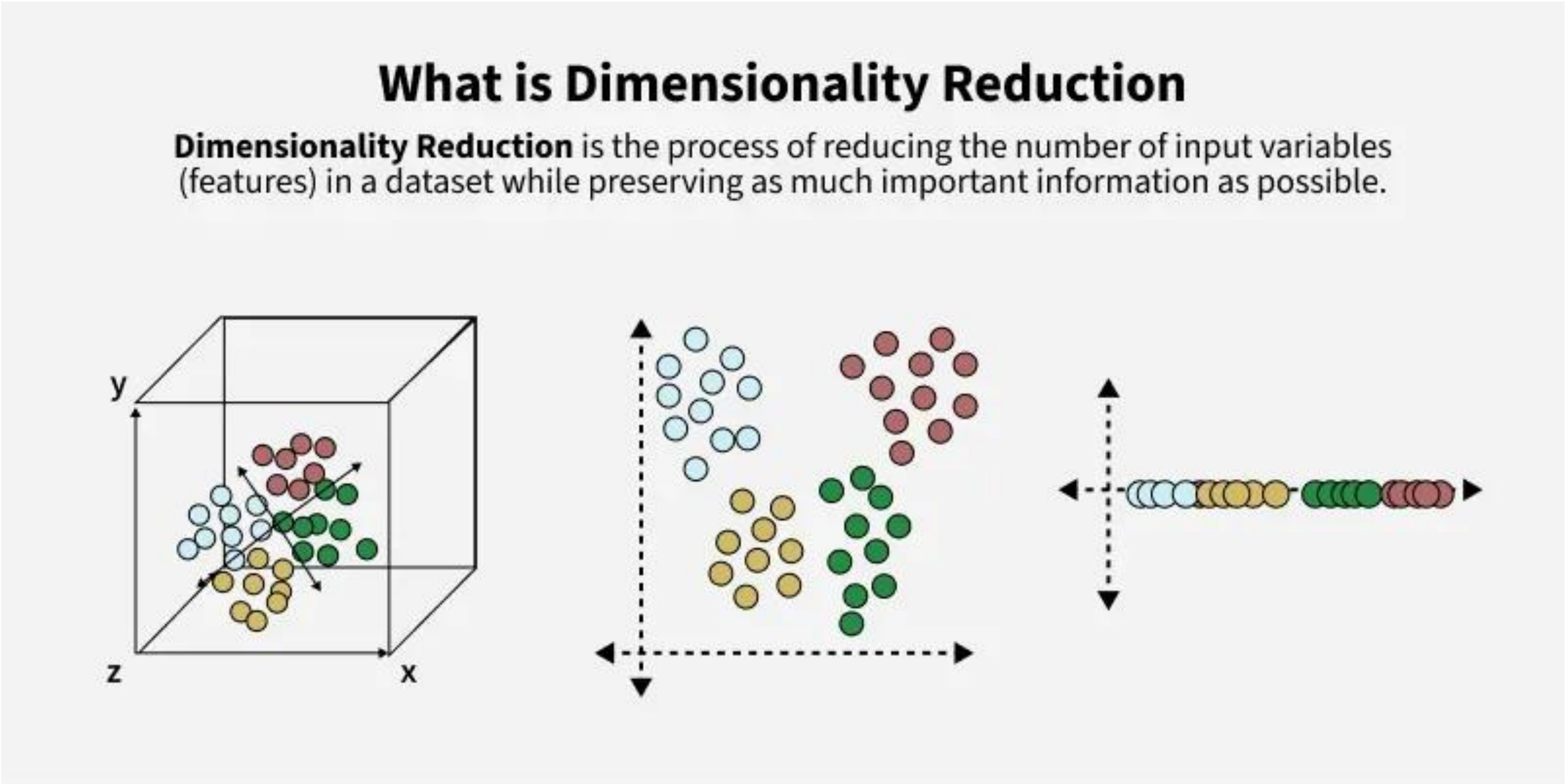
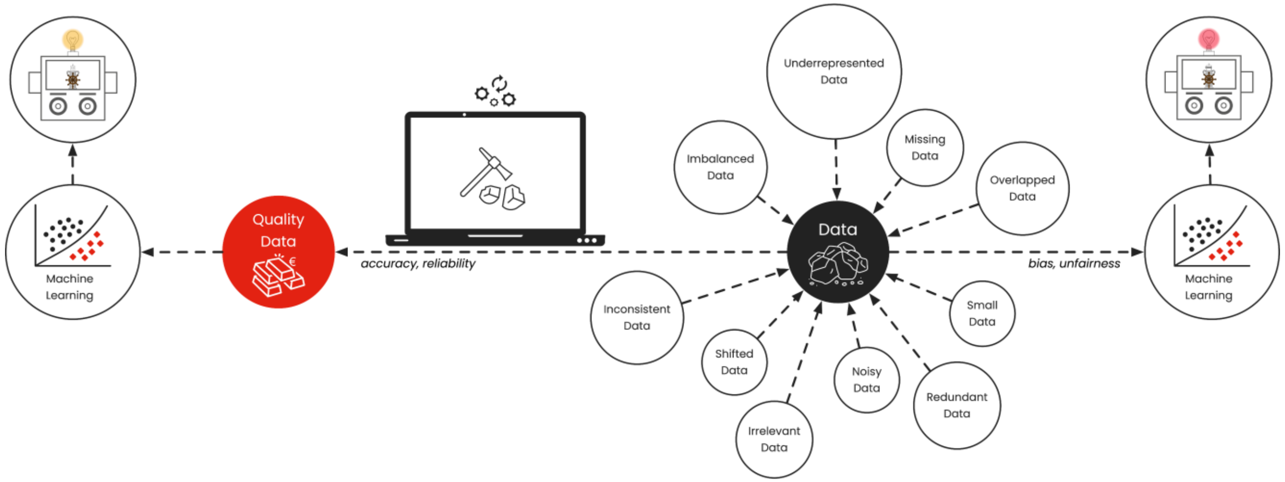


DIMENSIONALITY REDUCTION

Simplifying Complexity without Losing the Story

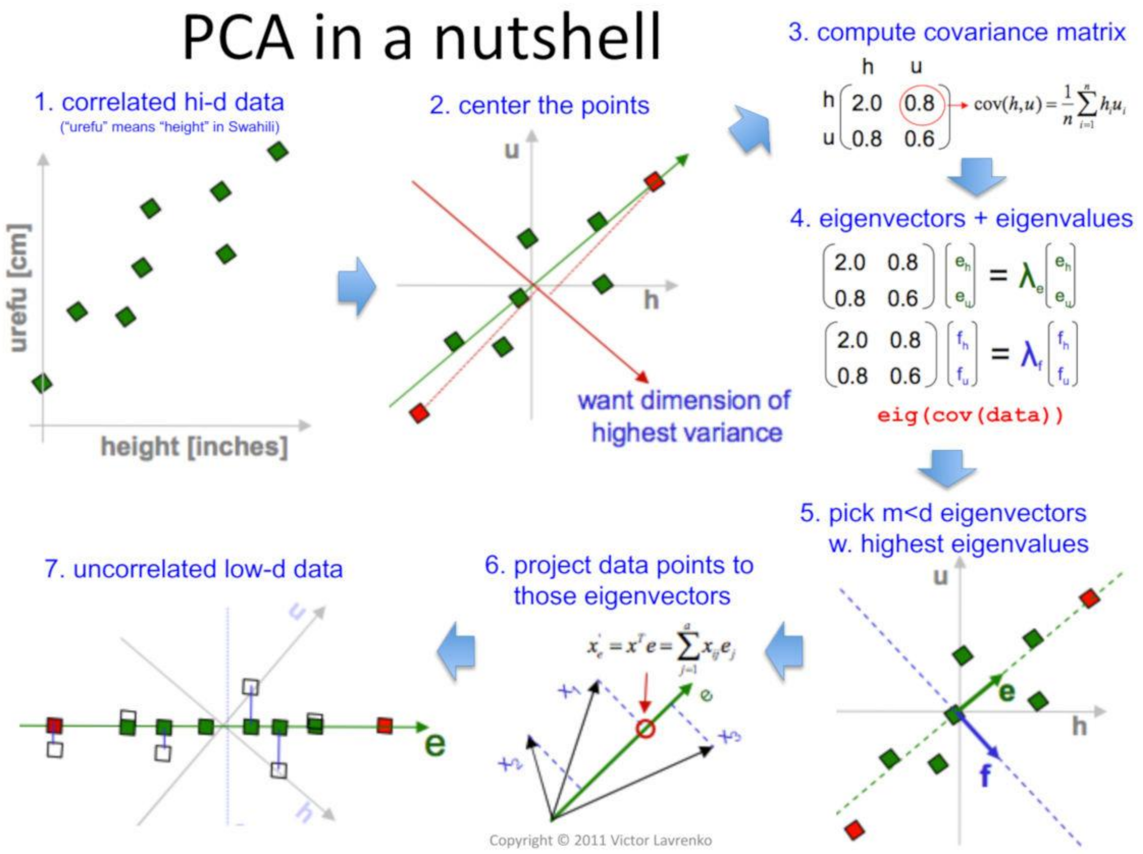
Instructor: Zaryab Ahmad Khan

THE PROBLEM OF TOO MUCH DATA

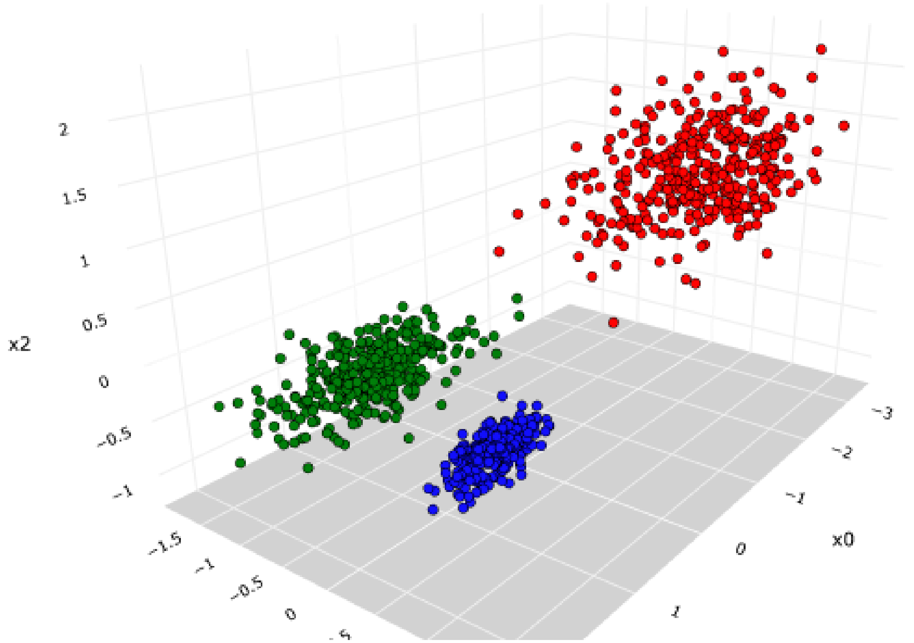


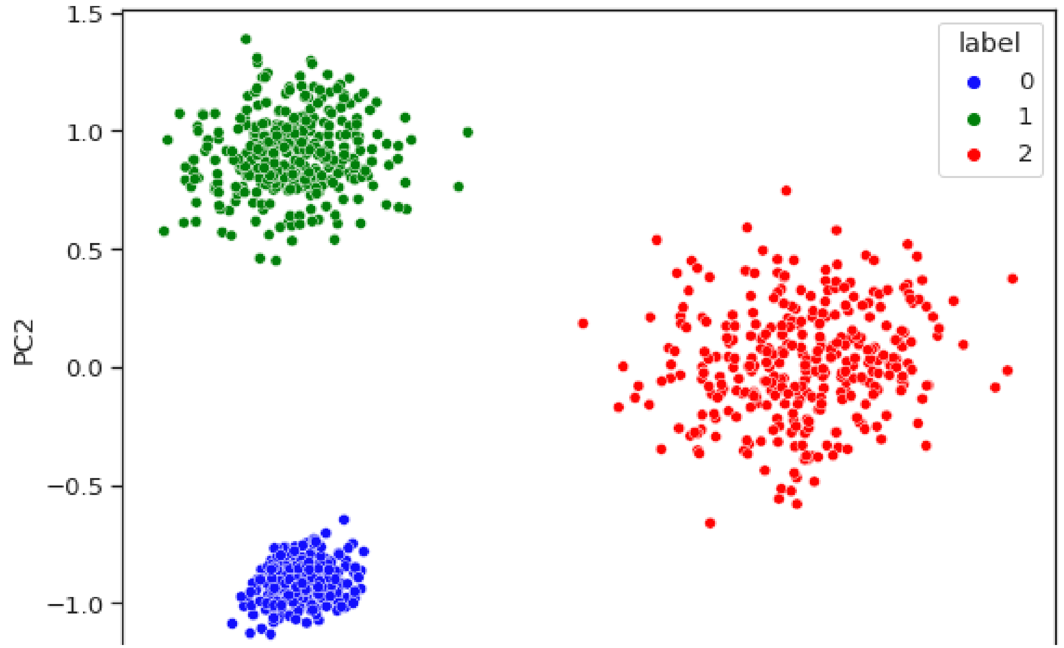
- **The "Curse of Dimensionality"**: Having 100 columns of data isn't always better.
- **Performance Hit**: Too many features can confuse an algorithm, increase training time, and actually make predictions *worse* by adding noise.
- **Analogy**: Trying to recognize a friend by looking at 1,000 microscopic skin cells (too much detail) vs. just looking at their face (the important "components").

PRINCIPAL COMPONENT ANALYSIS (PCA)



- **Definition**: A mathematical technique that safely compresses high-dimensional data (e.g., 20 columns) into just a few "Principal Components" (e.g., 3 columns).
- **Goal**: Keep the most important mathematical patterns and "variance" intact while throwing away the redundant fluff.





USING PCA FOR VISUALIZATION

- **The Human Limit**: Humans can visualize 2D (a flat chart) or 3D (a cube), but we cannot "see" 50 dimensions.
- **The Solution**: Use PCA to shrink complex data down to 2 dimensions so it can be plotted on a standard X/Y scatter chart.
- **Benefit**: This allows us to visually confirm if our data naturally forms clusters or groups.